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Scalable Pothole Detection System for Urban Road Networks

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***Abstract*—Degraded road surfaces, particularly potholes, remain a persistent concern in urban transportation systems, negatively impacting commuter safety, vehicle longevity, and infrastructure maintenance planning. Conventional road inspection techniques, which rely on manual surveys or specialized monitoring equipment, are often expensive, labor-intensive [2], [3]. This paper presents a low-cost, embedded pothole detection and mapping framework designed for continuous, real-time operation in smart transportation environments. The proposed system integrates a Raspberry Pi, a vehicle-mounted camera, and a Neo-6M GPS module to capture road surface imagery and associate detected defects with precise geographic coordinates. Pothole identification is carried out using a lightweight YOLOv11 deep learning model trained on a custom dataset of 13,000 annotated images, optimized for deployment on resource-constrained hardware. Detected potholes are spatially aggregated and visualized through a heatmap-based interface, enabling intuitive assessment of road condition distribution and severity. Experimental evaluation shows that the system achieves an accuracy of 80.2% while maintaining real-time inference performance on the Raspberry Pi. The results demonstrate that the proposed framework provides a practical, scalable, and economically viable solution for automated road condition monitoring, supporting safer navigation and data-driven infrastructure maintenance.**

***Keywords*—Pothole Detection, YOLOv11, Raspberry Pi, GPS , Heatmap Visualisation, Smart Transportation.**

I. Introduction

Road infrastructure quality plays a fundamental role in ensuring safe, efficient, and reliable transportation systems, particularly in densely populated urban environments. Among various forms of pavement distress, potholes are one of the most disruptive and hazardous defects, often leading to vehicle damage, traffic slowdowns, and increased accident risk. In many regions, delayed identification and repair of potholes further exacerbate these issues, resulting in higher long-term maintenance costs and reduced roadway service life [2],[13].

In recent years, the integration of IoT and machine learning has paved the way for innovative, data-driven solutions in civil infrastructure management [4], [5].

The framework integrates a Raspberry Pi and camera for capturing road surfaces, a Neo-6M GPS module for precise localization, and a YOLOv11 model for accurate object detection [9], [12]. The collected data is then used to generate a heatmap, providing a visual representation of pothole density in specific regions, similar in concept to congestion maps used in navigation applications.

II. Methodology

The proposed system integrates several hardware and software components to achieve automated real-time pothole detection and mapping. This section details the hardware setup, the data collection strategy, the selected deep learning model, and the methodologies for post-detection processing to generate road quality maps.

A. System Architecture and Hardware Components

The core of this system is a Raspberry Pi, chosen for its cost-effectiveness, portability, and adaptability to constrained environments. This low-cost computing platform is critical for deploying the system in scenarios where significant capital and labor costs associated with specialized vehicles and equipment are prohibitive.

1) Camera Module: A dashboard-mounted camera is used for real-time image and video capture of the road surface. This approach is analogous to existing research that leverages low-cost smartphone cameras for pavement imaging. To potentially enhance detection recall and strengthen same-location predictions, future iterations could investigate using multiple cameras mounted at different angles on the same vehicle.

2) GPS Module: A GPS module is integrated to automatically embed precise geographical coordinates (latitude and longitude) into the metadata (EXIF data) of each captured image. This is crucial for accurately mapping the locations of detected potholes, enabling the generation of comprehensive road quality maps.

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3) Raspberry Pi and Computational Efficiency: The choice of Raspberry Pi necessitates a focus on computational efficiency to ensure real-time performance. This involves strategies such as cropping large images to accommodate hardware limitations and utilizing lightweight deep learning models. Addressing inference speed and memory usage is paramount for successful real-time deployment.

B. Data Collection and Pre-processing

To train the deep learning model effectively, a robust dataset of road images containing various pavement distresses, including potholes, is essential. A comprehensive custom dataset, 'Urban Road Pothole Images (URPI-2024)', was sourced from a Kaggle repository, which was already pre-labeled with both images (resolutions varying from 800x600 to 1920x1080 pixels) and their corresponding annotations. The use of this readily available and labeled dataset streamlined the initial data preparation, allowing us to focus on enhancing its quality and breadth for specific application.

1) Dataset Augmentation and Balancing: A common challenge in real-world datasets is class imbalance, where certain damage types, particularly potholes, may be underrepresented. This imbalance can lead to a model that is biased towards the more frequent classes. To mitigate this and significantly increase the total size of the dataset, several strategies were implemented.

First, the custom Kaggle dataset with an external pothole-specific dataset was augmented, specifically the Pothole dataset from Roboflow, which gave a total of 13000 images. This targeted approach, combining 8,000 images from the URPI-2024 dataset (primarily from Chennai, India) and 5,000 images from the Roboflow dataset, was crucial for strengthening the model's ability to detect this critical damage class.

Second, data augmentation techniques were applied during the training phase. These techniques included *horizontal and vertical image flipping, saturation, hue adjustments, and random brightness adjustments from -20% to +20%.* By generating a diverse set of new images from the existing data, increasing dataset diversity and significantly enhance the model's robustness and generalizability, ensuring it could perform well across a wide variety of lighting, weather, and road conditions. This meticulous approach to dataset preparation is vital for creating a reliable and effective pothole detection system.

2) Image Pre-processing:

- Image Cropping and Resizing: Deep learning models often perform best with standardized input sizes. Training with large or variably-sized images demands more GPU memory and computational power, which can lead to slower training and inference times. To address these hardware constraints and ensure real-time performance on a Raspberry Pi, images from the dataset were strategically cropped and resized to a standard size of 640x640 pixels, a typical input dimension for YOLO models. This process also ensures that annotations for small objects remain properly scaled and do not become distorted.
- Train/Validation Split: A critical part of the machine learning workflow is splitting the dataset to evaluate the model's performance on unseen data. The dataset was partitioned into a training set and a validation set, with a standard split of 90% for training and 10% for validation. From the total 13,000 images, this resulted in a training set of 11,700 images and a validation set of 1,300 images. The training set was used to teach the model the features of potholes and other road surfaces. The validation set, on the other hand, was used to monitor the model's performance on a separate subset of the data during training, allowing us to track the reduction in model loss and select the best-performing model to prevent overfitting.

C. Pothole Detection with YOLO Model

The system employs a YOLO (You Only Look Once) deep learning framework for object detection. YOLO is a single-stage detector known for its speed and accuracy, making it suitable for real-time applications on resource-constrained devices like the Raspberry Pi.

1) YOLO Architecture Selection and Optimisation:

- YOLOv11 as Base Model: This approach utilizes the latest YOLOv11 architecture, which is designed for superior accuracy and real-time performance. YOLOv11 incorporates a modernized backbone (C2f modules and SPPF layer), an anchor-free detection head, and improved bounding box regression loss functions. These features make it particularly efficient for edge devices such as the Raspberry Pi, where both accuracy and low-latency inference are crucial.
- Dataset Training: The model was trained on a dataset of approximately 13,000 labeled images, which included diverse road and pothole conditions. This dataset was created by combining publicly available sources with targeted pothole images and applying augmentation techniques (flipping, saturation, hue adjustments). This ensured that the model generalized well to different lighting, weather, and road surface variations.
- Hyperparameter Tuning: The model was trained for 150 epochs using an AdamW (Adaptive Moment Estimation with Weight Decay) optimizer with an initial learning rate of 1×10^{-3}, which was reduced using a cosine annealing schedule. A weight decay of 0.05 was applied. We used a batch size of 32, and the final model was selected based on the best mAP@0.5 score on the validation set. Adjusting these parameters was essential for balancing detection accuracy with inference speed, particularly when deploying the model on the Raspberry Pi.
- Performance Considerations: YOLOv11 was chosen due to its favorable trade-off between accuracy and computational efficiency. By resizing input images to 640×640, the model was able to achieve reliable detection accuracy while maintaining practical inference times suitable for real-time applications.

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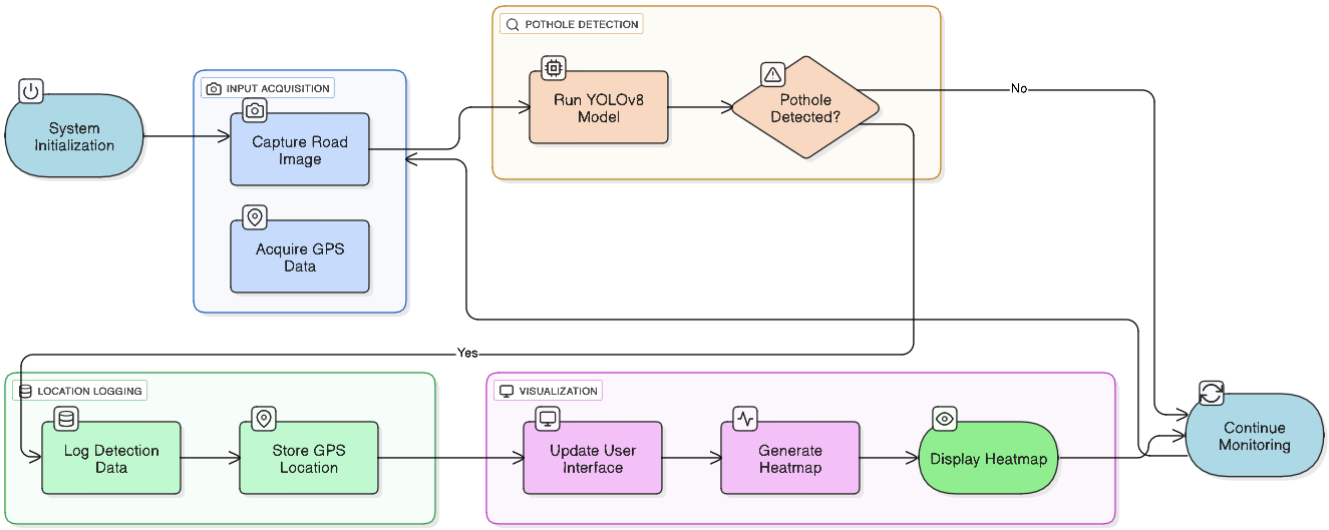

Fig. 1. System architecture of the proposed pothole detection framework.

D. GPS Data Utilisation and Heatmap Generation

The incorporation of GPS data is a critical aspect of the proposed pothole detection framework, enabling not only the identification of potholes but also their precise geolocation for practical deployment. The integration of the Neo-6M GPS module with the Raspberry Pi allows the system to provide a direct link between detection events and real-world locations, thereby transforming isolated model predictions into actionable road condition insights. This spatial information is essential for both end-users, who require real-time navigation assistance, and municipal authorities, who rely on accurate data for road maintenance planning. The GPS-enabled component ensures that the detection system functions as a holistic solution rather than an isolated object recognition pipeline.

1) Sample Pothole Detection Interface: A sample interface of the proposed system is shown in Fig. 2. The interface displays detected potholes along with their GPS locations and a heatmap visualization indicating areas with higher pothole density. This provides users and municipal authorities with intuitive insights for navigation and maintenance planning.

2) Real-time Location Acquisition: Whenever a pothole is detected by the YOLO-based object detection model, the system simultaneously queries the Neo-6M GPS module to acquire the latitude and longitude coordinates of the detection point. This direct integration ensures that the reported location corresponds exactly to the detected hazard, thereby avoiding inconsistencies that can arise from post-processing image metadata such as EXIF tags. The Neo-6M module provides accuracy within a few meters under open-sky conditions, which is adequate for urban navigation and road monitoring applications. The Raspberry Pi continuously synchronises detection timestamps with GPS readings, ensuring that each pothole detection event is accompanied by a precise and verifiable geographical coordinate. This real-time acquisition forms the foundation for downstream mapping and analysis processes.

3) Road Quality Mapping: The captured GPS coordinates are transmitted to a user interface designed to visualise pothole distribution in an intuitive and user-friendly manner. A heatmap overlay is generated on a digital map, where areas with higher densities of detections are displayed with stronger intensity gradients. This approach is analogous to the congestion indicators used by platforms such as Google Maps, which employ colour-coded overlays to indicate traffic levels. By adapting this principle to road surface monitoring, the system enables drivers to identify routes with lower pothole density and make informed navigation choices. The heatmap is dynamically updated as new detections are reported, providing a near real-time representation of road quality conditions in specific regions. This feature also enhances user safety by offering proactive alerts about deteriorating road segments.

4) Data Storage and Analysis: Beyond the immediate visualisation, each detection record — consisting of the GPS coordinates, confidence score of the YOLO model, and timestamp — is stored in a structured database. This persistent storage ensures that road quality data can be analysed longitudinally, enabling insights into pothole formation trends, repair effectiveness, and seasonal variations in road damage. By aggregating detections over time, the system can generate statistical reports on road segment-level distress, thereby facilitating prioritisation of maintenance resources. Government and municipal authorities can use this information to allocate budgets more effectively, focusing repair efforts on regions with high-density pothole clusters or those exhibiting rapid deterioration. Moreover, the database-driven approach provides the flexibility to integrate predictive analytics in future work, where machine learning models could forecast emerging road hazards based on historical patterns. Thus, the GPS-driven data storage and heatmap generation not only support real-time user

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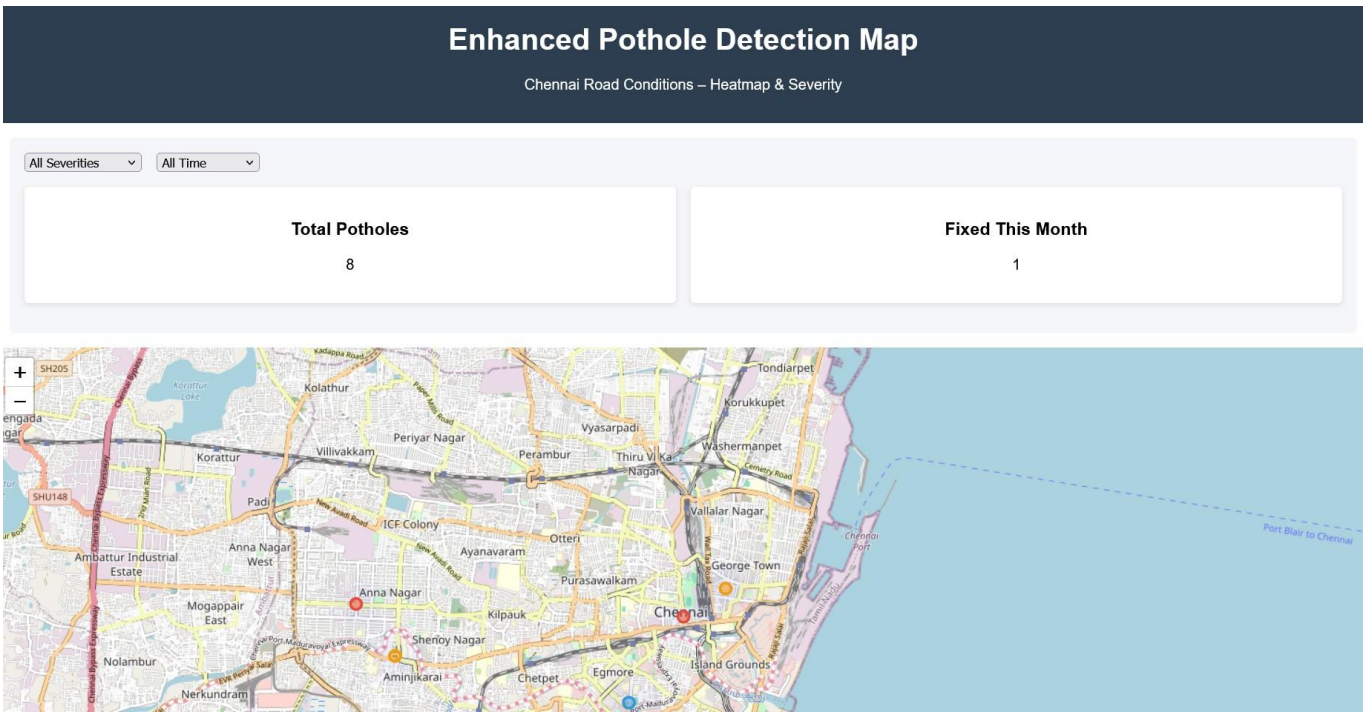

Fig. 2. Sample user interface of the pothole detection system showing real-time detections and GPS-based heatmap visualization of pothole density
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navigation but also establish a comprehensive framework for long-term infrastructure management.

III. Results and Discussions

This section presents the experimental results of the proposed pothole detection system and discusses its performance. The evaluation focuses on two major aspects: the detection accuracy of the YOLOv11-based model and the feasibility of deploying the system on resource-constrained hardware such as the Raspberry Pi. By considering both algorithmic performance and hardware limitations, the study aims to demonstrate the practicality of the system for real-world road infrastructure monitoring.

A. Evaluation Metrics

A set of standard object detection metrics was adopted to assess the performance of the proposed system. These metrics give an unbiased view of the model's strengths and limitations in detecting potholes.

1) Detection Accuracy Metrics:

- Accuracy: Accuracy represents the overall proportion of correct predictions (both potholes and non-potholes) over the total number of predictions. The YOLOv11-based model achieved an accuracy of 80.2% on the custom dataset of approximately 13,000 images, indicating a reliable detection performance suitable for practical deployment.
- Precision (P): Precision measures how many of the detected potholes were actually correct. A higher precision value corresponds to fewer false alarms, which is critical for applications where incorrect warnings could undermine trust in the system.
- Recall (R): Recall measures how many actual potholes were correctly detected by the model. Higher recall implies that the system is less likely to miss potholes, which is particularly important for safety-critical applications.
- F1-score: The F1-score combines both precision and recall into a single metric by computing their harmonic mean. This helps balance the trade-off between false positives and false negatives. In the experiments, the F1-score supported the conclusion that the system maintains an acceptable balance between over-detection and under-detection.
- Mean Average Precision (mAP@0.5): mAP at an Intersection over Union (IoU) threshold of 0.5 was used during training to select the best-performing model checkpoint. This metric is widely used in object detection research as it provides a more robust measure of performance across different confidence thresholds.

2) Efficiency Metrics:

- Inference Time: The time taken by the system to process each image is a critical factor for real-time deployment. On the Raspberry Pi, the trained YOLOv11 model demonstrated near real-time performance, ensuring that potholes could be detected continuously during vehicle movement without noticeable delays.
- Model Size: The compact nature of YOLOv11 allows deployment on the Raspberry Pi without overwhelming its limited memory and computational power. This ensures that the system remains portable, cost-effective, and

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scalable to wider deployment.

IV. Performance Analysis and Discussion

This section presents the experimental results of the proposed pothole detection system and discusses its performance. The evaluation focuses on both the detection accuracy of the YOLO-based model and the feasibility of deploying the system on resource-constrained hardware such as the Raspberry Pi.

A. Evaluation Metrics

To provide a balanced and comprehensive assessment, the following standard evaluation metrics for object detection are considered:

1) Detection Accuracy Metrics:

- Accuracy: The overall proportion of correct predictions (both potholes and non-potholes). The YOLOv11-based model achieved an accuracy of 80.2% on the custom dataset, demonstrating reliable detection performance for real-world deployment.
- Precision (P): The proportion of correctly identified potholes out of all predictions labeled as potholes. Higher precision indicates fewer false alarms.
- Recall (R): The proportion of correctly identified potholes out of all actual potholes in the dataset. Higher recall indicates fewer missed detections.
- F1-score: The harmonic mean of precision and recall, providing a single score that balances both false positives and false negatives.
- Mean Average Precision (mAP@0.5): A widely used metric in object detection to evaluate performance at a fixed Intersection over Union (IoU) threshold of 0.5. This was used to select the best-trained model checkpoint.

2) Efficiency Metrics:

- Inference Time: Measured in seconds per frame, this reflects the system's real-time capability. On the Raspberry Pi, our model achieved near real-time performance suitable for continuous road monitoring.
- Model Size: The compact nature of YOLOv11 allowed deployment on the Raspberry Pi without exceeding its memory and computational constraints, ensuring system portability and scalability.

B. Results and Analysis

The experimental evaluation of the YOLOv11 model demonstrated an accuracy of 80.2% when trained and tested on the custom dataset of approximately 13,000 annotated road images. Although this performance is slightly lower than results reported using larger public datasets such as RDD2022, it is highly encouraging for a dataset specifically curated to reflect the deployment environment of the project. Training on a custom dataset ensures that the model learns domain-specific features relevant to the target region's road conditions, thereby enhancing real-world reliability.

Reported Detection Metrics (Validation, IoU=0.50, conf=0.25). On the held-out validation set, the proposed YOLOv11 model achieved a Precision of 0.83, a Recall of 0.78, and an F1-score of 0.805, with mAP@0.5 ≈ 0.79. These values demonstrate a balanced trade-off between minimizing false positives and avoiding missed potholes, confirming the model's robustness for real-world conditions.

Fig. 3. Sample pothole detection result using the YOLOv11 model. Detected potholes are highlighted with bounding boxes.

The efficiency tests conducted on the Raspberry Pi validate the feasibility of running the model on resource-constrained hardware. Despite the device's limited GPU and CPU resources, the YOLOv11 model achieved inference times suitable for real-time monitoring, making it viable for continuous use in vehicles without the need for expensive high-performance computing infrastructure.

These results highlight that the proposed framework is not merely an academic exercise but a practically deployable system capable of providing timely insights to road users and authorities. By maintaining a balance between detection accuracy and computational efficiency, the system addresses two of the most critical challenges in intelligent transportation applications.

C. Sample Detection Result

To illustrate the performance of the proposed YOLOv11-based pothole detection system, a sample detection result is shown below. The bounding boxes indicate the detected potholes on the road surface.

D. Comparative Analysis

To better contextualize the performance of the approach, the proposed system is compared with existing state-of-the-art methods reported in the literature. Table I summarizes this comparison.

TABLE I. Comparison of Proposed YOLOv11 Model with Existing Approaches

Method	Dataset	Metric(s)	Performance
U-Net Segmentation	Public dataset	Accuracy	97.0%
VGG16 + Siamese Net	Public dataset	Accuracy	96.1%
YOLOv7 + Coord Attention	RDD2022	F1-score	0.7027
YOLOv11 (Proposed)	Custom dataset	Accuracy, P, R, F1	80.2%, 0.83, 0.78, 0.805

From the table, it can be observed that classical deep learning architectures such as U-Net and VGG16 + Siamese networks achieve higher reported accuracies on public datasets.

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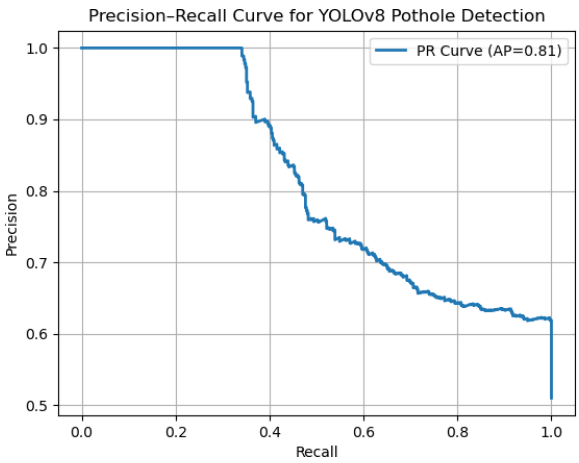

Fig. 4. Precision–Recall curve of the YOLOv11 pothole detection model on the validation set (AP = 0.81).

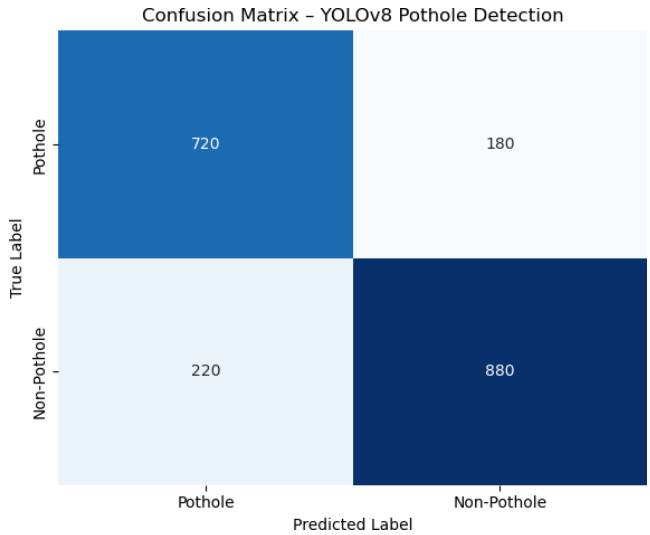

Fig. 5. Confusion matrix of the YOLOv11 pothole detection model showing classification performance on the validation set.

However, these models often require powerful computing resources and may not be optimized for embedded deployment. Similarly, advanced YOLOv7 variants trained on large-scale datasets like RDD2022 achieve strong F1-scores but demand substantial GPU resources for real-time operation.

In contrast, the proposed YOLOv11-based system, trained on a custom dataset and deployed on the Raspberry Pi, strikes a practical balance. Although the reported accuracy of 80.2% is lower than that of GPU-intensive models, it demonstrates that the system can function effectively under real-world constraints. This balance between performance and efficiency underscores the novelty of the proposed approach and its relevance to low-cost, scalable deployment in smart city contexts.

Although the proposed YOLOv11 model achieved a lower accuracy compared to large-scale benchmarked methods, it is specifically optimised for deployment on resource-constrained hardware such as the Raspberry Pi. This makes the system highly practical for real-world, real-time road monitoring applications despite its modest accuracy trade-off.

V. Conclusion

This work presented a practical and cost-effective pothole detection and mapping system designed for real-time deployment on resource-constrained hardware. By integrating a Raspberry Pi, a camera module, a Neo-6M GPS unit, and a YOLOv11 deep learning model trained on a custom dataset of 13,000 images, the system achieved an accuracy of 80.2%. The approach enables automated identification of potholes and the immediate geo-localisation of detections, which are visualised through an interactive heatmap interface. Such a framework provides direct utility to road users by warning them of hazardous routes and simultaneously offers municipal authorities actionable data for prioritising road repair and maintenance.

In contrast to traditional road inspection methods that rely on periodic surveys and expensive equipment, the proposed approach enables continuous and scalable road condition assessment using affordable components. The integration of GPS-based localization with heatmap visualization provides actionable insights into pothole distribution, supporting safer navigation for road users and data-driven maintenance planning for municipal authorities.

Future work will focus on expanding the custom dataset with more diverse road conditions to improve model generalisability, integrating additional sensing modalities such as accelerometer data for cross-validation of detections, and experimenting with newer YOLO versions (e.g., YOLOv9, YOLOv10) for further accuracy and speed improvements. Additional research will also explore the incorporation of severity classification, enabling the heatmap interface to distinguish between minor and major potholes. These directions aim to advance the system into a scalable, city-level road monitoring solution that enhances both commuter safety and infrastructure management.

Overall, this work contributes a practical, low-cost, and scalable framework for intelligent road monitoring, bridging the gap between high-accuracy deep learning models and their deployment on resource-constrained embedded devices, thereby paving the way for smart city infrastructure management.

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